

Using ML, DL, NLP

①

Trust Models etc

Sec 39 PCA

lec-3 PCA Geometric
Intuition

lec-4 PCA Maths
Intuition

lec-5 Eigen decomposition
on covariance
matrix

Housing	Dataset	Model
size of house	no. of rooms	



$P \propto A$

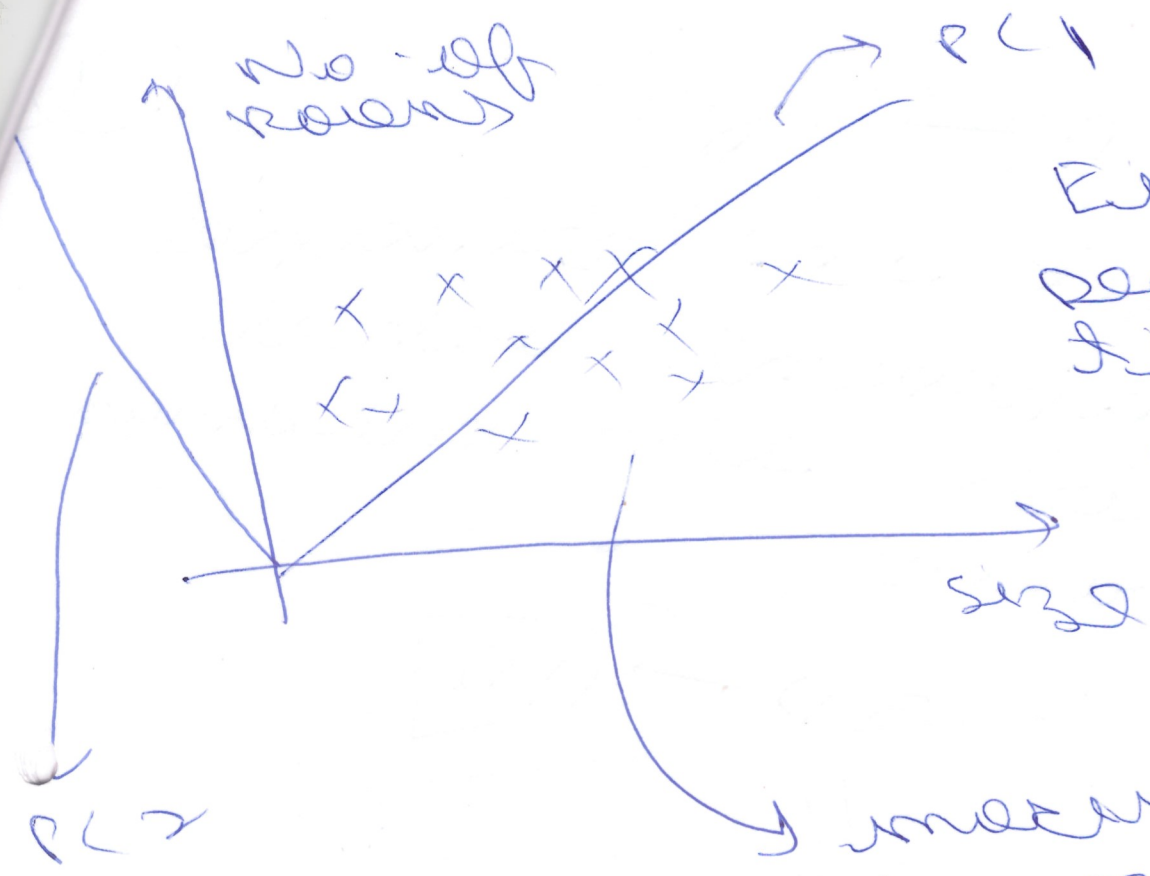
2 dimension $\rightarrow$ 1 dimension



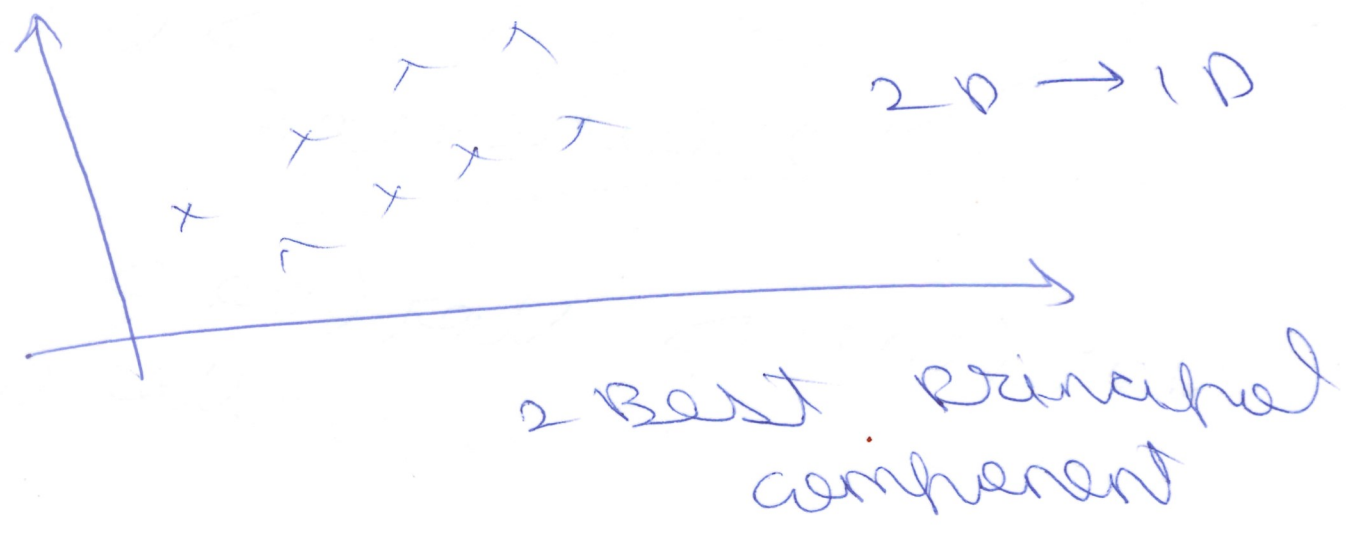
$2D \rightarrow 1D$

(+) Less of information (Not)

3



2D $\rightarrow$ 1D Dimension
much information is not lost

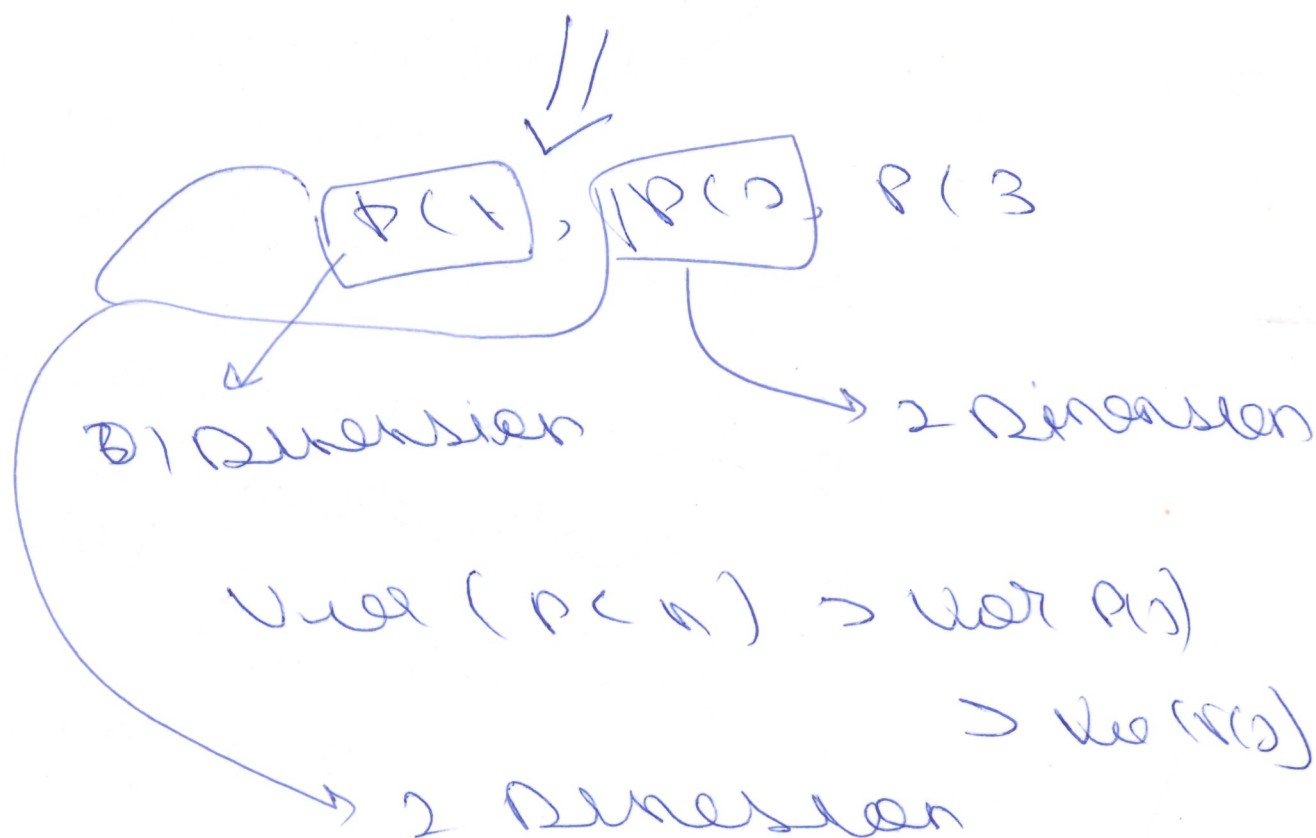


3D $\rightarrow$ 1D

(4)

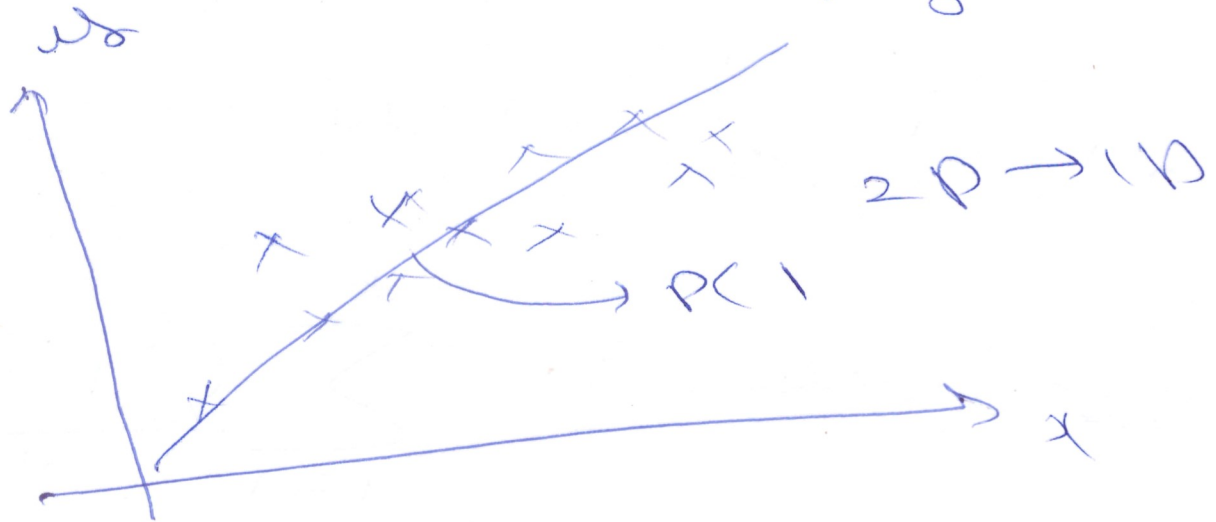
To get the principal
component which
captures the maximum
variance.

3D $\rightarrow$ 1D



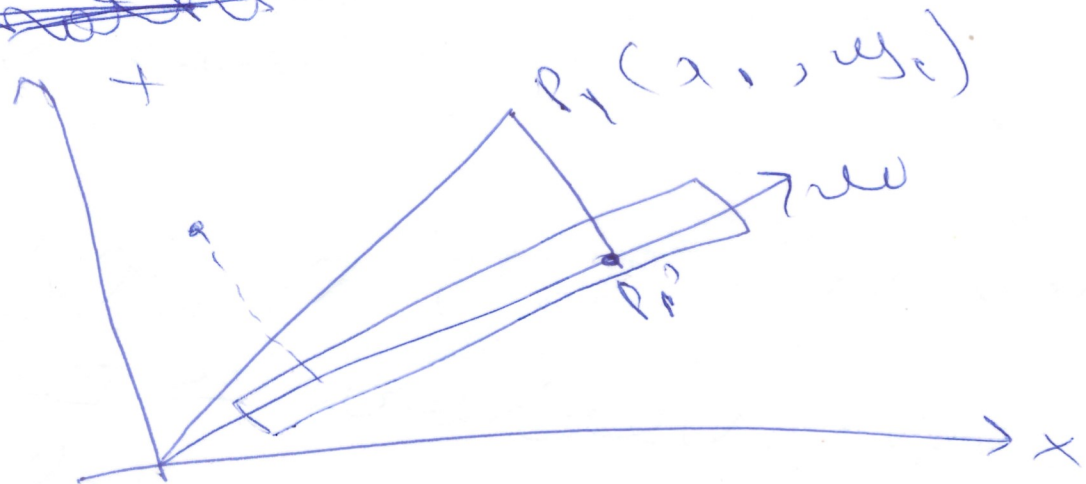
$\text{Var}(P(1)) > \text{Var}(P(2)) > \text{Var}(P(3))$

$\Rightarrow$ metric intuition ~~behind~~ $P(A)$ (5)
~~behind~~ $P(A)$
 Algorithm



- (1) projection
- (2) cost function

~~$\Rightarrow$ ~~metric intuition~~~~



$$\text{Proj}_{p, u} = \frac{p_1 \cdot u}{\|u\|}$$

$$\text{Proj}_{p, u} = p_1 \cdot u$$

$P_1 \rightarrow P_2$

(6)

$P_0', P_1', P_2', P_3', P_4', \dots, P_s'$

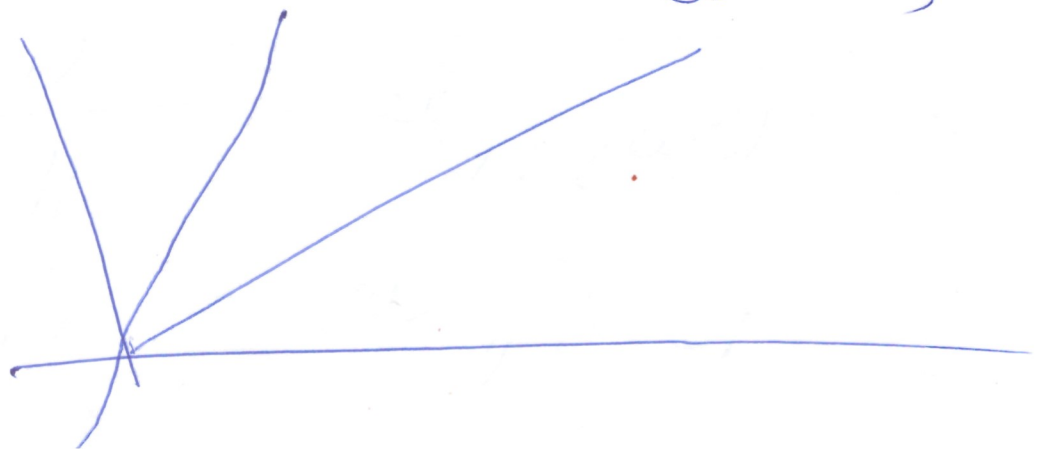
$x_0', x_1', x_2', x_3', x_4', \dots, x_s'$

$$\text{variance} = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n}$$

[Goal $\equiv$ Find
the best
unit vector
which captures
max. variance

$P_0 \rightarrow P_1', P_2', P_3', P_4' \dots$

Eigen Vectors $\rightarrow$ Eigen
values



① Covariance matrix b/w features

② Eigen vectors & Eigen values will be found out from this covariance

③ Eigen vector $\rightarrow$ Eigen value $\rightarrow$ magnitude of

Eigen vector $\rightarrow$ capture max. variance

Linear transformation of matrix

$$A v = \lambda v$$

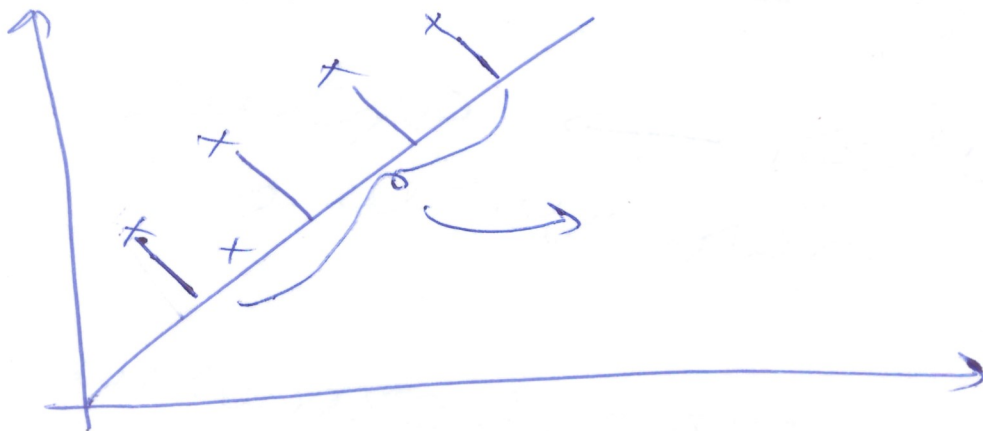
Eigen vectors & Eigen
vectors

[Linear transform
matrix]

[Eigen decomposition of
square matrix]



Eigen vector &
Eigen values



$$\begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} \times \begin{bmatrix} v \end{bmatrix} = \lambda * v$$

Eigen values

2) A is my matrix

$$A * v = \lambda A v$$

This equation is used for calculating eigen values $\rightarrow$ eigen vectors



Whichever Eigen vector has max magnitude, that will be used as a flag or principal component.

$\Rightarrow$ It will be able to capture maximum variance

Eigen vector $\rightarrow$ max magnitude



max Eigen vector



Best Principal Component $\Rightarrow PC1$

Steps to calculate Eigen values & Eigen vectors

① Covariance of Features

$$\begin{bmatrix} x & y \end{bmatrix} \quad 2$$

$\downarrow$
 x_1

$$\text{Cov}(x, y) =$$

$$\Rightarrow \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

2x2

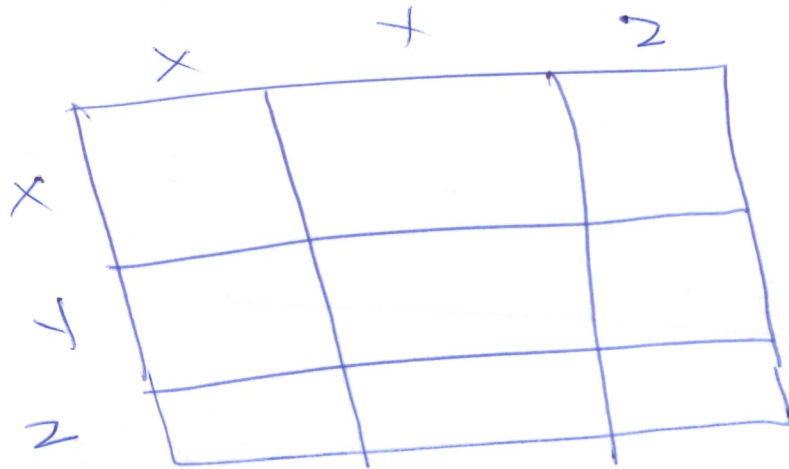
	x	y
x	$\text{var}(x)$	$\text{Cov}(x, y)$
y	$\text{Cov}(y, x)$	$\text{var}(y)$

$$\text{Cov}(x, x) = \text{var}(x)$$

$$\text{Cov}(x, x) = \text{var}(x)$$

$$\text{Cov}(y, y) = \text{var}(y)$$

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$$A \cdot v = \lambda v$$

$$| \theta_1 \quad \theta_2 \quad \theta_3 |$$

$$A \cdot v = \lambda \cdot v$$

$$\lambda_1, \lambda_2$$

Eigen values

$$PC_1, PC_2$$

$$\lambda_1$$

$$PC_1$$

$$\lambda_2$$

$$PC_2$$

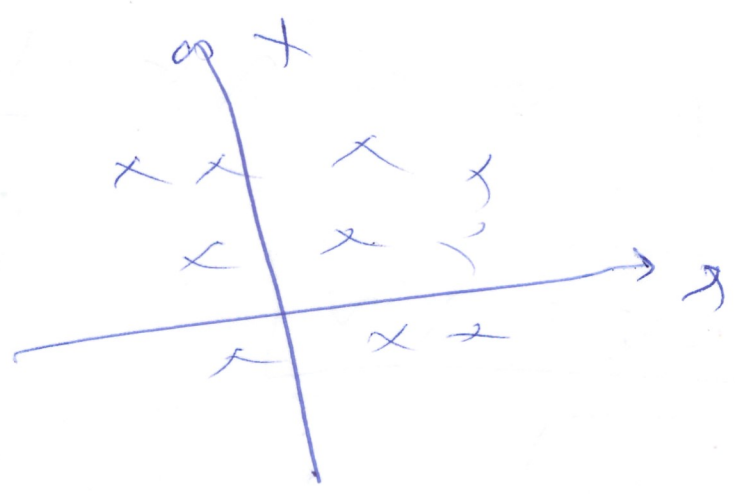


(*)



standardize the data

(1)



(2) covariance matrix of x and y

x	$\text{var}(x)$	$\text{cov}(x, y)$
y	$\text{cov}(y, x)$	$\text{var}(y)$

3) Find Eigen values over $\mathbb{R}$

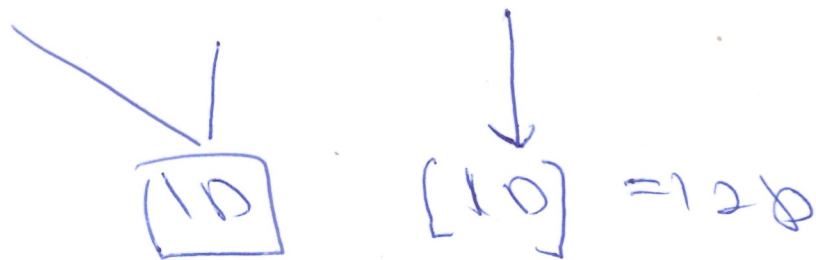
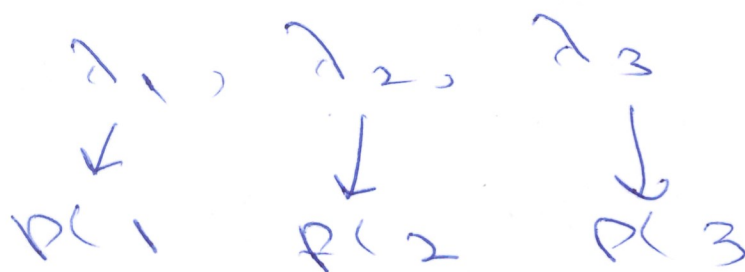
(13)

$$A v = \lambda v$$

$\boxed{\lambda_1, \lambda_2} \Rightarrow$ Eigen values



$$3D \rightarrow 1D$$



$$[1D] = 120$$

$$3D \rightarrow 1D$$

$$\lambda_1, \lambda_2, \lambda_3$$



$$2D \rightarrow 1D$$

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$$\lambda_1 \rightarrow \lambda_2$$



$P(1)$



$D(2)$



1D

$\Rightarrow$ projection